



NORTH SYDNEY BOYS HIGH SCHOOL

2009 YEAR 12 HSC ASSESSMENT TASK 2

Mathematics

Extension 2

General Instructions

- Working time – **55 minutes**
- Write on the lined paper in the booklet provided
- Write using blue or black pen
- Board approved calculators may be used
- All necessary working should be shown in every question
- Each new question is to be started on a **new page**.

- Attempt all questions

Class Teacher:

(Please tick or highlight)

- ☐ Mr Barrett
- ☐ Mr Fletcher
- ☐ Mr Weiss

Student Number:

Question No	1	2	3	4	Total	Total
Mark	$\overline{13}$	$\overline{9}$	$\overline{11}$	$\overline{7}$	$\overline{40}$	$\overline{100}$

QUESTION 1 (13 marks)**Marks**

The ellipse E has equation $\frac{x^2}{4} + \frac{y^2}{3} = 1$

- (i) Calculate the eccentricity 1
- (ii) Write down the coordinates of the foci S and S' 2
- (iii) Write down the equation of each directrix 2
- (iv) Sketch E, showing all essential features 2
- (v) Show that the tangent at the point P (x_1, y_1) on the ellipse is
$$\frac{xx_1}{4} + \frac{yy_1}{3} = 1$$
 3
- (vi) The tangent at P intersects the directrix at the point T, where T has coordinates (m,n) and $m > 0$. Show that PT subtends a right angle at S. 3

QUESTION 2 (9 marks) Start a new page

- (a) Use the method of mathematical induction to show that 4
 $4^n > 2n + 1$ where n is a positive integer.
- (b) Resolve the following into partial fractions :
 - (i) $\frac{2}{(x-1)(x+3)}$ 2
 - (ii) $\frac{4x+2}{(x+3)(x^2+1)}$ 3

QUESTION 3 (11 marks) [Start a new page](#)**Marks**

- (i) Sketch the graph of the hyperbola $9x^2 - 16y^2 = 144$,
showing clearly the foci, the directrices, and the asymptotes. 4
- (ii) Show that the normal at P ($4\sec\theta$, $3\tan\theta$) has equation 3
- $$\frac{4x}{\sec\theta} + \frac{3y}{\tan\theta} = 25$$
- (iii) A line through P parallel to the y-axis meets the asymptote 2
in the first quadrant at Q. The normal at P meets the x-axis at G.
Find the coordinates of Q and G.
- (iv) Show that GQ is perpendicular to OQ. 2

QUESTION 4 (7 marks) [Start a new page](#)

The points P ($2p, \frac{2}{p}$) and Q ($2q, \frac{2}{q}$) lie on the rectangular hyperbola $xy = 4$

M is the midpoint of PQ. P and Q move on the hyperbola such that the chord PQ always passes through the point (4, 2).

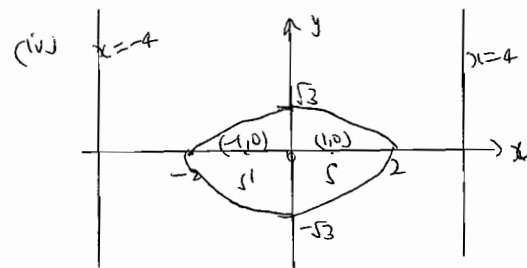
- (i) Show that PQ has equation $x + pqy = 2 (p + q)$ 3
- (ii) Show that $pq = p + q - 2$ 1
- (iii) Find the equation of the locus of M. 3

SOLUTIONS TO 4 UNIT ASSESSMENT 2

Q1 (i) $a=2, b=\sqrt{3}$
 $e^2 = 1 - \frac{b^2}{a^2}$
 $= \frac{1}{4}$
 $\therefore e = \frac{1}{2}$

(ii) S is (1,0)
 S' is (-1,0)

(iii) $x=4$
 $x=-4$



(v) $\frac{x^2}{4} + \frac{y^2}{3} = 1$
 Differentiate both sides with respect to x

$$\Rightarrow \frac{2x}{4} + \frac{2y}{3} \cdot \frac{dy}{dx} = 0$$

$$\frac{dy}{dx} = -\frac{3x}{4y}$$

At P, $\frac{dy}{dx} = -\frac{3x_1}{4y_1}$

Equation of tangent at P: $y - y_1 = -\frac{3x_1}{4y_1}(x - x_1)$

$$4yy_1 - 4y^2 = -3xx_1 + 3x_1^2$$

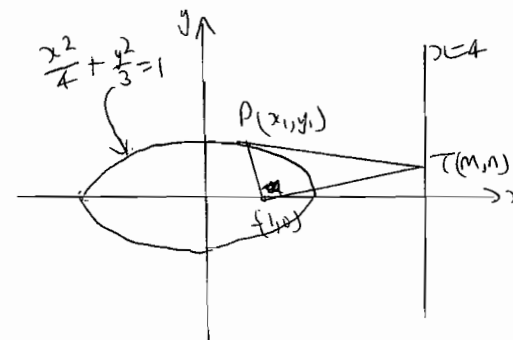
$$3xx_1 + 4yy_1 = 3x_1^2 + 4y_1^2$$

$$\frac{xx_1}{4} + \frac{yy_1}{3} = \frac{x_1^2}{4} + \frac{y_1^2}{3}$$

But $\frac{x_1^2}{4} + \frac{y_1^2}{3} = 1$ because P(x₁, y₁) lies on ellipse

$$\therefore \frac{xx_1}{4} + \frac{yy_1}{3} = 1$$

(vi)



To find coordinates of T
 sub. $x=4$ into equation of PT

$$\Rightarrow x_1 + \frac{yy_1}{3} = 1$$

$$y = \frac{3}{y_1}(1 - x_1)$$

i.e. T is $(4, \frac{3}{y_1}(1 - x_1))$

gradient SP = $\frac{y_1}{x_1 - 1}$

gradient PT = $\frac{3}{y_1} \cdot \frac{(1 - x_1)}{3} = \frac{1 - x_1}{y_1}$

gradient SP $\times$ gradient PT = $\frac{y_1}{x_1 - 1} \times \frac{1 - x_1}{y_1}$
 $= -1$

$\therefore SP \perp PT$

-1 for each missing feature

Q2 (a) When $n=1$, LHS = $4^1 = 4$
 RHS = $2 \times 1 + 1 = 3$
 $LHS > RHS$
 $\therefore$ true for $n=1$

Assume true for $n=k$, where k is a positive integer

i.e. $4^k > 2k+1$

Try to prove true for $n=k+1$

i.e. required to prove $4^{k+1} > 2(k+1)+1$

i.e. $4^{k+1} > 2k+3$

proof: $4^{k+1} = 4 \times 4^k > 4(2k+1)$ from assumption

i.e. $4^{k+1} > 8k+4$
 $> 2k+3$ because $k > 0$

$\therefore$ true for $n=k+1$ if true for $n=k$

But already proved true for $n=1$

$\therefore$ true for $n=2, 3, 4, \dots$ etc

and $\therefore$ true for all positive integers n

(b) (i) $\frac{2}{(x-1)(x+3)} = \frac{a}{x-1} + \frac{b}{x+3}$

$2 = a(x+3) + b(x-1)$

sub. $x=1 \Rightarrow 4a=2 \Rightarrow a=\frac{1}{2}$

sub. $x=-3 \Rightarrow -4b=2 \Rightarrow b=-\frac{1}{2}$

then $\frac{2}{(x-1)(x+3)} = \frac{1}{2(x-1)} - \frac{1}{2(x+3)}$

(ii) $\frac{4x+2}{(x+3)(x^2+1)} = \frac{a}{x+3} + \frac{bx+c}{x^2+1}$

$4x+2 = a(x^2+1) + (x+3)(bx+c)$

sub. $x=-3 \Rightarrow 10a=-10 \Rightarrow a=-1$

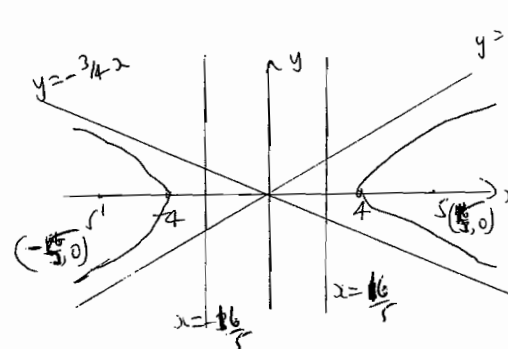
equating coefficients of $x^2 \Rightarrow a+b=0 \Rightarrow b=1$

equating constants $\Rightarrow a+3c=2 \Rightarrow c=1$

then $\frac{4x+2}{(x+3)(x^2+1)} = \frac{-1}{x+3} + \frac{x+1}{x^2+1}$

Q3

(i)



4

-1 for each part missing

$\frac{x^2}{16} - \frac{y^2}{9} = 1$ $a=4, b=3$
 $e = \sqrt{1 + \frac{b^2}{a^2}} = \frac{5}{4}$

or

$9x^2 - 16y^2 = 144$
 Differentiate both sides with respect to x

$\Rightarrow 18x - 32y \frac{dy}{dx} = 0$

$\frac{dy}{dx} = \frac{18x}{32y} = \frac{9x}{16y}$

gradient of normal at P

$= -\frac{16y_1}{9x_1}$

$= -\frac{4 \tan \theta}{3 \sec \theta}$

$= -\frac{4 \tan \theta}{3 \sec \theta}$

(ii) $x = 4 \sec \theta$

$\frac{dx}{d\theta} = 4 \sec \theta \tan \theta$

$y = 3 \tan \theta$

$\frac{dy}{d\theta} = 3 \sec^2 \theta$

$\frac{dy}{dx} = \frac{\frac{dy}{d\theta}}{\frac{dx}{d\theta}} = \frac{3 \sec^2 \theta}{4 \sec \theta \tan \theta}$

$= \frac{3 \sec \theta}{4 \tan \theta}$

gradient of normal at P = $-\frac{4 \tan \theta}{3 \sec \theta}$

equation of normal at P is

$y - 3 \tan \theta = -\frac{4 \tan \theta}{3 \sec \theta} (x - 4 \sec \theta)$

$3y \sec \theta - 9 \tan \theta \sec \theta = -4x \tan \theta + 16 \sec \theta \tan \theta$

$4x \tan \theta + 3y \sec \theta = 25 \tan \theta \sec \theta$

$\frac{4x}{\sec \theta} + \frac{3y}{\tan \theta} = 25$

or
 equation of normal at (x_1, y_1) is

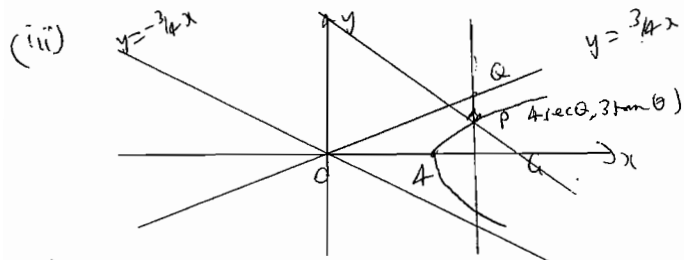
$y - y_1 = -\frac{16y_1}{9x_1} (x - x_1)$

$9x_1 y - 9x_1 y_1 = -16x y_1 + 16x_1 y_1$

$(6x y_1 + 9y x_1 = 25 x_1 y_1)$

$\frac{16x}{x_1} + \frac{9y}{y_1} = 25$

sub. $x_1 = 4 \sec \theta, y_1 = 3 \tan \theta$



To find coordinates of Q: sub $x = 4 \sec \theta$ into $y = \frac{3}{4}x$
 $\Rightarrow y = 3 \tan \theta$
 i.e. Q is $(4 \sec \theta, 3 \tan \theta)$

To find coordinates of A: sub $y = 0$ into $\frac{4x}{\sec \theta} + \frac{3y}{\tan \theta} = 25$
 $\Rightarrow x = \frac{25 \sec \theta}{4}$
 i.e. A is $(\frac{25}{4} \sec \theta, 0)$

(iv) gradient OQ = $\frac{3}{4}$
 gradient AQ = $\frac{-3 \tan \theta}{\frac{25}{4} \sec \theta} = \frac{-12}{25} = -\frac{4}{5}$
 gradient OQ $\times$ gradient AQ = -1
 $\therefore OQ \perp AQ$

Q4

(i) gradient PQ = $\frac{\frac{2}{q} - \frac{2}{p}}{2q - 2p} = \frac{\frac{p-q}{pq}}{q-p} = -\frac{1}{pq}$

equation PQ: $y - \frac{2}{p} = -\frac{1}{pq}(x - 2p)$

$pqy - 2q = -x + 2p$

i.e. $x + pqy = 2(p+q)$

(ii) PQ passes through (4, 2)

sub $x=4, y=2 \Rightarrow 4 + 2pq = 2(p+q)$

$2 + pq = p+q$

i.e. $pq = p+q-2$

(iii) coords. of M are $(\frac{2p+2q}{2}, \frac{\frac{2}{p}+\frac{2}{q}}{2})$
 $= (p+q, \frac{1}{p} + \frac{1}{q})$

$x = p+q \quad \dots (1)$

$y = \frac{1}{p} + \frac{1}{q} \quad \dots (2)$

$y = \frac{q+p}{pq} = \frac{p+q}{p+q-2}$

i.e. $y = \frac{x}{x-2}$